

Jason Bandy

Software Engineering Leader | Platform Engineering | AI/ML | Enterprise Mobile Systems

Nashville, TN | 615-840-0718 | jason.bandy948@gmail.com | linkedin.com/in/jason-bandy-11bba650

EXECUTIVE SUMMARY

Software engineering and platform leader with 15+ years of experience building developer platforms, mobile systems, AI-powered automation, and high-performing engineering organizations. Currently leading systems and technology programs for Amazon Operations Technology Solutions, combining hands-on technical architecture with global execution across 400K+ devices and 1,200+ facilities. Known for turning complex operational problems into scalable software, data, and platform solutions that reduce cost, improve reliability, and accelerate decision-making.

TECHNICAL SKILLS

Languages & Data: Python, SQL, JavaScript / TypeScript, Java, Swift, Kotlin, pandas, SQLite, PostgreSQL

Platform & DevOps: Backstage, Terraform, CI/CD, Jenkins, GitLab, Docker, Kubernetes, API management, observability

Cloud & AI: AWS Bedrock, Claude, AWS Lambda, S3, Google Cloud Platform, LLM integration, prompt engineering, NLP

Mobile Systems: Android SDK, iOS, MDM, barcode scanning systems, Bluetooth peripherals, device fleet management

PROFESSIONAL EXPERIENCE

Amazon - Operations Technology Solutions
System Development Manager | Nashville, TN

October 2023 - Present

AI/ML Platform Development and Automation

- Built an AI-powered capital planning analytics platform supporting **\$250M+ in investment decisions**, integrating Python, SQLite, pandas, AWS Bedrock, and Claude to analyze large-scale equipment and device datasets.
- Designed a natural language query interface that translates business questions into SQL, allowing non-technical stakeholders to explore complex equipment data without waiting on manual analysis.
- Implemented a multi-iteration optimization engine with constraint validation, dynamic code execution, feedback loops, and scenario comparison for device allocation across **800+ facilities**.
- Created Excel-to-SQLite ETL pipelines with automated schema inference, data validation, and reporting in Markdown, CSV, JSON, HTML, and PNG.
- Reduced capital planning analysis cycles from **weeks to hours**, enabling faster scenario planning for strategic sourcing and lifecycle decisions.

Enterprise Mobile Systems and Technical Architecture

- Designed a universal Bluetooth scanner abstraction strategy enabling cross-vendor scanner compatibility across Zebra, Honeywell, and PointMobile ecosystems.
- Architected provider-based scanner interfaces for Android applications, reducing hardcoded vendor dependencies and enabling any supported ring scanner to work with any supported host device.
- Delivered immediate cross-compatibility fixes for scanner source validation, unblocking ring scanner deployments across EU, APAC, and Americas regions.
- Deployed Device Fleet Health Check APIs with five production endpoints for warehouse inventory, battery health aggregation, emergent battery events, time-series capacity degradation, and unhealthy battery thresholds across a **400K+ device fleet**.

Strategic Technology Programs and Operational Execution

- Led a **\$205M mobile technology modernization program** replacing **386K+ devices** across **1,200+ global facilities**, moving from single-vendor dependency to a function-based multi-vendor strategy.
- Delivered **\$92M+ in savings** and a **32% per-unit cost reduction** through competitive sourcing, device optimization, and vendor diversification.
- Built financial and technical models projecting **\$126M annual productivity gains**, a 14-month payback period, 83% IRR, and \$455M five-year NPV.
- Exceeded a Q4 2025 cost-savings goal by delivering **\$14.9M**, reaching **131% of target** and reducing Zebra dependency from **80.4% to 75.96%**.
- Certified successor devices for five end-of-sale product lines months ahead of schedule, protecting fulfillment operations from emergency procurement and support-risk scenarios.

National Federation of Independent Business (NFIB)

August 2023 - October 2023

National Director, Software Engineering | Nashville, TN

- Led modernization of software development practices with a DevOps-focused operating model.
- Defined 1-, 3-, and 5-year engineering roadmap aligning technical delivery to business outcomes.
- Established engineering standards, tooling strategy, and development process improvements across teams.

HCA Healthcare

May 2019 - July 2023

Senior Director, Software Engineering Center | Nashville, TN

- Built and led an enterprise Engineering Center responsible for developer experience, cloud governance, engineering standards, and platform enablement.
- Established Developer Platform teams delivering internal tooling, reusable services, CI/CD templates, and shared DevEx capabilities for product engineering teams.
- Implemented Backstage developer portal strategy to unify access to tools, services, documentation, and engineering standards.
- Led Cloud Architecture and Governance, including Terraform standards, cloud architecture patterns, security alignment, cost visibility, and FinOps practices.
- Re-architected engineering team structures using Team Topologies, shifting from functional silos to platform, enabling, and stream-aligned team models.

HCA Healthcare

July 2017 - May 2019

Principal Mobile Software Engineer

- Led enterprise mobile strategy across iOS and Android platforms, including mobile architecture, release governance, secure authentication, offline-capable architecture, and mobile DevOps practices.

HCA Healthcare

June 2016 - July 2017

Senior Staff Mobile Engineer

- Led front-end mobile development for enterprise nurse-leader rounding applications and built end-to-end CI/CD pipelines for iOS and Android build, test, and release automation.

SELECTED TECHNICAL ACHIEVEMENTS

AI-Powered Capital Planning Platform: Production-grade Python and AWS Bedrock tooling for natural language analysis, optimization, and reporting.

Universal Scanner Compatibility: Cross-vendor scanner abstraction and compatibility patterns for enterprise Android devices.

Device Fleet Health APIs: Real-time API foundations for battery health, event detection, lifecycle management, and predictive maintenance.

Developer Experience Platform: Backstage, CI/CD templates, DevOps practices, and engineering playbooks for enterprise teams.

MEASURABLE BUSINESS IMPACT

- **\$250M+** technology and capital planning portfolio supported through technical strategy, analytics, and execution.
- **\$92M+** savings delivered through device optimization, competitive sourcing, and multi-vendor strategy.
- **\$175M** capital expenditure avoidance through lifecycle timing, portfolio optimization, and procurement strategy.
- **400K+** devices and **1,200+** global facilities impacted through modernization, deployment, monitoring, and lifecycle programs.

EDUCATION AND CERTIFICATIONS

Middle Tennessee State University - Bachelor of Science, Computer Science | 2009 - 2013

Certifications: FinOps Certified Practitioner (FOCP) | Google Cloud Digital Leader | Backstage: Developer Portals